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Evaluate Yourself

8. Critical temperature of H_2O, NH_3, and CO_2 are 647.4, 405.5 and 304.2 K, respectively. When we start cooling from a temperature of 700 K which will liquefy first and which will liquefy finally?

6.6.1 Derivation of critical constants from van der Waals constant:

The van der Waals equation for n moles is

$$\left(P + \frac{a\,n^2}{V^2}\right)(V - nb) = nRT \quad \text{---------- (6.22)}$$

For 1 mole

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT \quad \text{---------- (6.23)}$$

From the equation we can derive the values of critical constants P_c, V_c and T_c in terms of a and b, the van der Waals constants, On expanding the above equation

$$PV + \frac{a}{V} - Pb - \frac{ab}{V^2} - RT = 0 \text{ ------- (6.24)}$$

Multiply equation (6.24) by V^2 / P

$$\frac{V^2}{P}\left(PV + \frac{a}{V} - Pb - \frac{ab}{V^2} - RT\right) = 0$$

$$V^3 + \frac{aV}{P} + -bV^2 - \frac{ab}{P} - \frac{RTV^2}{P} = 0 \text{ --- (6.25)}$$

When the above equation is rearranged in powers of V

$$V^3 - \left[\frac{RT}{P} + b\right]V^2 + \left[\frac{a}{P}\right]V - \left[\frac{ab}{P}\right] = 0 \text{ --- (6.26)}$$

The equation (6.26) is a cubic equation in V. On solving this equation, we will get three solutions. At the critical point all these three solutions of V are equal to the critical volume V_C. The pressure and temperature becomes P_c and T_c respectively

i.e., $V = V_C$

$$V - V_C = 0$$

$$(V - V_C)^3 = 0$$

$$V^3 - 3V_C V^2 + 3V_C{}^2 V - V_C{}^3 = 0 \quad \text{---- (6.27)}$$

As equation (6.26) is identical with equation (6.27), we can equate the coefficients of V^2, V and constant terms in (6.26) and (6.27).

$$-3V_C V^2 = -\left[\frac{RT_C}{P_C} + b\right]V^2$$

$$3V_C = \frac{RT_C}{P_C} + b \quad \text{---- (6.28)}$$

$$3V_C^2 = \frac{a}{P_C} \quad \text{---- (6.29)}$$

$$V_C^3 = \frac{ab}{P_C} \quad \text{---- (6.30)}$$

Divide equation (6.30) by equation (6.29)

$$\frac{V_C^3}{3V_C^2} = \frac{ab / P_C}{a / P_C}$$

$$\frac{V_C}{3} = b$$

i.e. $V_C = 3b$ ------- (6.31)

when equation (6.31) is substituted in (6.29)